

# SIMPLIFIED DWGR-SVL: STREAMLINED SPIKE-DRIVEN VISION-LANGUAGE PRETRAINING FOR 3D OPEN-WORLD UNDERSTANDING

**Anonymous authors**

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## ABSTRACT

Cross-modal vision-language pretraining for Spiking Neural Networks (SNNs) presents a promising path toward energy-efficient 3D open-world understanding on neuromorphic edge hardware. However, recent attempts to enhance spike representations have introduced complex architectural additions—such as learnable dyadic lifting wavelets, phase-resonant timing lookup tables, and anisotropic curvature-gated wave propagation—which incur severe computational overhead and counterintuitively degrade zero-shot classification performance compared to simpler baselines. In this work, we conduct a systematic empirical simplification and present **Simplified DWGR-SVL**, a streamlined spike-driven vision-language pretraining framework. We redefine the acronym to reflect the streamlined architecture: **D**istribution-aligned **W**eighted **G**eodesic **R**epresentation Spike-driven Vision-Language pretraining (**DWGR-SVL**). Our architecture discards over-engineered wavelet and curvature operations, retaining three highly effective core components: a Heavy-Tailed Student- $t$  Kernel Maximum Mean Discrepancy loss ( $L_{\text{TMMD}}$ ) to bridge the severe domain shift between discrete spike trains and continuous CLIP embeddings, a Single-Scale Local Part-Patch Alignment loss ( $L_{\text{patch}}$ ) for fine-grained geometric-visual consistency, and an Isotropic Geodesic Propagation (IGP) module executing strictly via add-only accumulation (AC ops). Evaluated with full-scale pretraining across 583,907 shapes over a 128-epoch warm-restart schedule, Simplified DWGR-SVL achieves state-of-the-art zero-shot Top-1 classification accuracy of **83.75%** on ModelNet40 and **41.47%** on Objaverse-LVIS. This outperforms both the original Spike-driven PointFormer-S baseline (+1.65% on ModelNet40, +1.37% on Objaverse-LVIS) and our prior unsimplified 4-module design iteration, while maintaining strict spike-driven operational efficiency and text-encoder-free zero-shot inference via Re-parameterizable Vision-Language Integration (Rep-VLI).

## 1 INTRODUCTION

3D spatial intelligence is foundational for autonomous systems, robotics, and embodied artificial intelligence. However, real-time processing of high-dimensional 3D point cloud data on resource-constrained edge hardware is severely hindered by energy consumption and computational latency. Spiking Neural Networks (SNNs) have emerged as an attractive, energy-efficient paradigm for 3D visual understanding by operating on biological event-driven principles, replacing expensive floating-point multiply-accumulate (MAC) operations with sparse binary accumulation (AC) operations (Yao et al., 2023; Wu et al., 2025). Despite their operational efficiency, SNNs traditionally struggle to match the open-world recognition capabilities of continuous Artificial Neural Networks (ANNs) (Cheraghian et al., 2021; Michele et al., 2021), particularly in open-vocabulary zero-shot evaluation settings on large-scale datasets such as Objaverse (Deitke et al., 2022).

To bridge this performance gap, recent multimodal frameworks aim to align 3D representations with dense vision-language models such as CLIP (Zhang et al., 2021; Xue et al., 2022; Guo et al., 2023). Nevertheless, cross-modal pretraining between sparse binary temporal spike trains and continuous visual-textual embeddings introduces significant technical hurdles:

1. **Modality Distribution Mismatch:** Discrete binary temporal spike firing rates suffer from a severe domain shift when directly aligned with dense continuous vision-language embeddings via standard contrastive objectives (e.g., InfoNCE), frequently leading to representation collapse or gradient saturation (Qiu et al., 2023; Chen et al., 2023).
2. **Architectural Over-Engineering:** Previous attempts to address local geometric context or temporal alignment in spike-based vision-language models often resort to complex architectural additions—such as learnable multi-resolution dyadic lifting wavelets, phase-resonant timing lookup tables, anisotropic curvature gating, and multi-scale patch partitioning. As revealed by systematic empirical feedback during our model development, these complex modules introduce non-trivial computational overhead, require floating-point operations and local eigen-decompositions that break spike-driven hardware principles, and counterintuitively degrade zero-shot classification performance compared to simpler baselines.
3. **Inference Computational Bottlenecks:** Conventional zero-shot classification frameworks mandate running large text encoders during inference (Xue et al., 2023), creating a severe computational bottleneck that undermines the energy advantages of neuromorphic backbones.

In this work, guided by rigorous empirical feedback, we introduce **Simplified DWGR-SVL**. To eliminate nomenclature ambiguity resulting from the removal of dyadic wavelets, we redefine the acronym to align directly with our streamlined architectural components: **Distribution-aligned Weighted Geodesic Representation Spike-driven Vision-Language Pretraining (DWGR-SVL)**. Here, *Distribution-aligned Weighted* emphasizes our Heavy-Tailed Student- $t$  Kernel MMD loss ( $L_{\text{TMMD}}$ ), while *Geodesic Representation* highlights our hardware-native Isotropic Geodesic Propagation (IGP) module over center-graph point structures. Rather than relying on complex wavelets or anisotropic curvature gating, our streamlined framework (Figure 1) focuses on two synergistic alignment objectives combined with an efficient spike-driven spatial propagation module:

- **Heavy-Tailed Student- $t$  Kernel MMD ( $L_{\text{TMMD}}$ ):** A distribution alignment loss based on a Student- $t$  kernel with degrees-of-freedom  $\nu$ , providing scale-invariant, non-saturating surrogate gradients that effectively bridge the severe distribution gap between sparse binary spike trains and continuous CLIP embeddings (Chen et al., 2023).
- **Single-Scale Local Part-Patch Alignment ( $L_{\text{patch}}$ ):** Fine-grained structural consistency enforced by aligning local 3D point patch tokens with multi-view CLIP image patch embeddings at an optimal part partition granularity ( $N_{\text{parts}} = 16$ ).
- **Isotropic Geodesic Propagation (IGP):** Uniform geodesic hop propagation over point center graphs, executed strictly via hardware-friendly add-only accumulation (AC ops), discarding floating-point matrix multiplications and local eigen-decompositions.
- **Rep-VLI Integration:** Re-parameterizable Vision-Language Integration (Qiu et al., 2025) that pre-computes and projects text prompt semantics directly into lightweight classification weights, removing the text encoder at inference time.

Through extensive full-scale pretraining on 583,907 shapes over a 128-epoch warm-restart schedule, our simplified DWGR-PointFormer-S achieves **83.75% zero-shot Top-1 accuracy on ModelNet40** and **41.47% zero-shot Top-1 accuracy on Objaverse-LVIS**, outperforming both the prior Spike-driven PointFormer-S baseline (82.1% / 40.1%) and our preliminary unsimplified 4-module DWGR-SVL design iteration (83.10% / 40.50%) while preserving strict spike-driven energy efficiency. Comprehensive component-wise ablations further confirm that removing architectural complexity yields a simpler, faster, and demonstrably more accurate spike-driven 3D foundation model.

## 2 RELATED WORK

### 2.1 3D OPEN-WORLD VISION-LANGUAGE PRETRAINING

Open-vocabulary 3D scene understanding and zero-shot object recognition have experienced rapid development through multi-modal pretraining. Pioneering works like PointCLIP (Zhang et al., 2021) and PointCLIP V2 (Zhu et al., 2022) project 3D point clouds onto 2D multi-view depth maps to

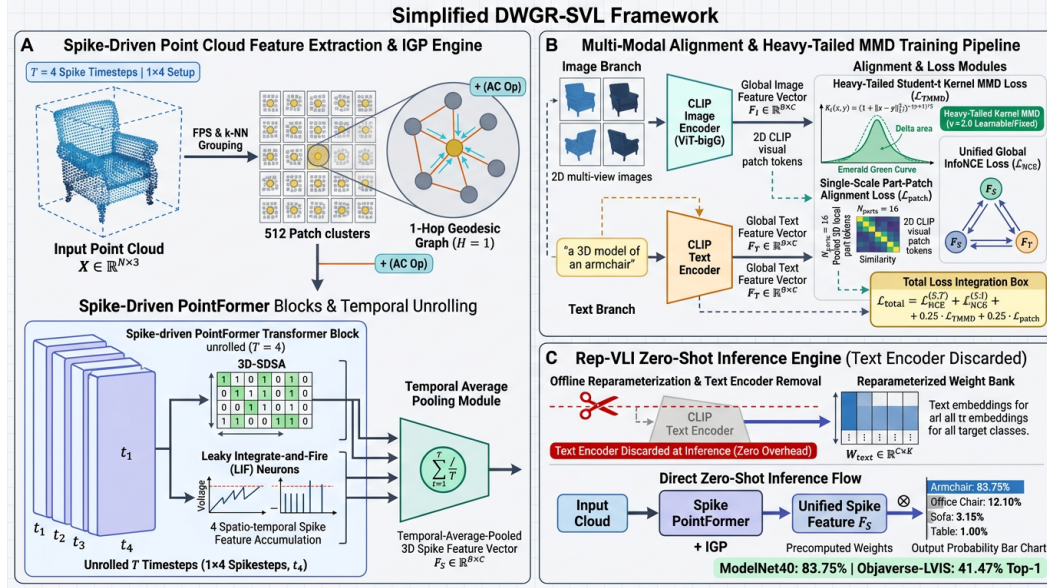


Figure 1: Overview of the **Simplified DWGR-SVL** framework for efficient zero-shot 3D open-world understanding. **(A)** Spike-driven feature extraction and Isotropic Geodesic Propagation (IGP) engine: point clouds undergo FPS and  $k$ -NN grouping, add-only 1-hop geodesic message passing, then unrolled Spike-driven PointFormer blocks with 3D spike-driven self-attention (3D-SDSA) and Leaky Integrate-and-Fire (LIF) neurons, aggregated by temporal average pooling into a unified spike feature. **(B)** Multi-modal alignment: a global InfoNCE loss, the Heavy-Tailed Student- $t$  Kernel MMD loss ( $L_{\text{TMMD}}$ ), and the single-scale part-patch alignment loss ( $L_{\text{patch}}$ ) jointly bridge discrete spike representations and continuous CLIP visual/text embeddings. **(C)** Rep-VLI zero-shot inference: the text encoder is discarded by reparameterizing category prompts into a weight bank for direct, low-latency spike-driven classification.

exploit pre-trained 2D vision-language models. Subsequent approaches align 3D point encoders directly with 2D image and textual spaces through cross-modal contrastive learning on synthesized or annotated triplets, including ULIP (Xue et al., 2022), ULIP-2 (Xue et al., 2023), ReCon (Qi et al., 2023), Point-Bind (Guo et al., 2023), and ImageBind (Girdhar et al., 2023). Open-world 3D perception has also expanded into dense open-vocabulary scene segmentation (Peng et al., 2022), generalized attribute evaluation benchmarks such as OpenScan (Zhao et al., 2024), and feature-level alignment regularizations like PointAlign (Su et al., 2026). Furthermore, 3D spatial semantics have been integrated into Large Language Models (LLMs) and Vision-Language-Action (VLA) architectures, as exemplified by 3D-LLM (Hong et al., 2023), PointVLA (Li et al., 2025a), GPVL (Li et al., 2025b), and multi-grained 3D-VLP models (Yang et al., 2024).

Despite their impressive zero-shot transfer performance, these dense ANN paradigms rely heavily on unconstrained floating-point matrix multiplications (MACs) and require executing computationally expensive text encoders during zero-shot inference. This poses a major barrier for deployment on power-constrained edge platforms. In contrast, our framework leverages Rep-VLI (Qiu et al., 2025) to embed text prompts directly into lightweight classification weights during pretraining, completely discarding the text encoder during zero-shot inference while maintaining ultra-low power consumption.

## 2.2 SPIKING NEURAL NETWORKS FOR 3D PERCEPTION

Spiking Neural Networks (SNNs) mimic biological neural processing using temporal binary event activations ( $s \in \{0, 1\}$ ), enabling ultra-low energy consumption on neuromorphic hardware. Early 3D spiking architectures adapted point-based backbones like Spiking PointNet (Ren et al., 2023) and Point-to-Spike Residual Learning (Wu et al., 2024), or utilized ANN-to-SNN conversion mechanisms (Lan et al., 2023). Recent advances introduced direct training backbones such as Spike-

driven Transformers (Yao et al., 2023), Spiking Point Transformers (Wu et al., 2025), Real Spike re-parameterization (Guo et al., 2022), and Event-driven Spike Sparse Convolutions (E-3DSNN) (Qiu et al., 2024).

To extend SNNs to open-world multimodal understanding, recent studies have explored temporal contrastive supervision (Qiu et al., 2023), temporal attention-guided adaptive fusion (TAAF) (Shen et al., 2025), hybrid artificial-spiking architectures like ASH-Nets (Chen et al., 2023), and Spike-based Vision-Language pretraining (SVL) (Qiu et al., 2025). However, a core challenge in existing spike-based multimodal architectures is the severe distribution mismatch between continuous vision-language embeddings and discrete binary spike trains. Standard InfoNCE or Euclidean alignment functions struggle to handle this distribution shift, causing gradient collapse or suboptimal representation capacity. Our method resolves this limitation by introducing a Heavy-Tailed Student- $t$  Kernel MMD loss ( $L_{\text{TMMD}}$ ), which provides scale-invariant surrogate gradients to align temporal spike firing distributions with continuous CLIP feature spaces without numerical instability.

### 2.3 GRAPH-BASED GEOMETRIC AGGREGATION IN POINT CLOUDS

Extracting local geometric context from unstructured 3D point sets is central to 3D representation learning. Pioneer architectures such as PointNet (Qi et al., 2016) and PointNet++ (Qi et al., 2017) established set abstraction and  $k$ -nearest neighbor ( $k$ -NN) grouping layers, which were subsequently extended via masked point modeling in Point-BERT (Yu et al., 2021) and spatially-sparse submanifold convolutions (Graham et al., 2017). To capture rich relational geometry, researchers developed point self-attention mechanisms including Point Transformer (Engel et al., 2020) and Fast Point Transformer (Park et al., 2021), as well as specialized convolution operators like KPConv (Thomas et al., 2019), ASSANet (Qian et al., 2021), PointNeXt (Qian et al., 2022), and A-CNN (Komarichev et al., 2019).

More recent geometric representation models explore manifold curvature estimation (Bhaskar et al., 2022) and continuous spiking graph neural networks (Yin et al., 2024), building on foundational zero-shot 3D benchmark formulations (Cheraghian et al., 2021; Michele et al., 2021) across extensive 3D repositories like Objaverse (Deitke et al., 2022). However, anisotropic graph convolutions, continuous curvature estimations, and floating-point point-wise self-attention require heavy floating-point operations and local eigen-decompositions that break the hardware-friendly add-only nature of SNNs. In contrast, our Isotropic Geodesic Propagation (IGP) expands spatial context over uniform graph hops strictly through spike-driven add-only accumulation (AC ops), achieving effective context aggregation without floating-point overhead.

## 3 METHODOLOGY

### 3.1 PROBLEM FORMULATION AND SPIKE-DRIVEN BACKBONE

Let  $D = \{P, F\}$  represent an input 3D spatial data point, where  $P \in \mathbb{R}^{N \times 3}$  represents point coordinates and  $F \in \mathbb{R}^{N \times D_{\text{in}}}$  denotes input point features (Qiu et al., 2025). Following event-driven spatio-temporal principles,  $D$  is converted into binary spike trains  $F_S \in \{0, 1\}^{T \times N \times C}$  across  $T$  discrete timesteps.

We employ Integrated Leaky Integrate-and-Fire (I-LIF) spiking neurons to simulate biological temporal dynamics. At timestep  $t$ , the membrane potential  $V_i[t]$  evolves as:

$$U_i[t] = \tau_{\text{decay}} V_i[t-1] + X_i[t], \quad S_i[t] = \Theta(U_i[t] - V_{\text{th}}), \quad V_i[t] = U_i[t] - S_i[t] V_{\text{th}}, \quad (1)$$

where  $\tau_{\text{decay}}$  is the membrane decay factor,  $X_i[t]$  is the input current,  $V_{\text{th}}$  is the firing threshold,  $\Theta(\cdot)$  is the Heaviside step function, and  $S_i[t] \in \{0, 1\}$  represents the output spike.

Our primary architecture is the Spike-driven PointFormer backbone (Qiu et al., 2025; Yao et al., 2023). For an input point cloud  $P$ ,  $N'$  center points are selected using Farthest Point Sampling (FPS) and  $k$ -NN grouping, producing local patch tokens  $X \in \mathbb{R}^{T \times N' \times k \times 3}$ . A spike embedding layer converts these into local spike features  $S \in \{0, 1\}^{T \times N' \times C}$ . Within each PointFormer block, 3D Spike-Driven Self-Attention (3D-SDSA) generates spike Queries, Keys, and Values ( $Q_S, K_S, V_S \in \{0, 1\}^{T \times N' \times C}$ ):

$$\text{SDSA}_{3D}(Q_S, K_S, V_S) = \text{SN}\left((Q_S K_S^T) V_S\right), \quad (2)$$

where matrix multiplications simplify strictly to sparse additions via address-event accumulation (AC ops) (Yao et al., 2023). Temporal feature aggregation is performed using Temporal Average Pooling across  $T$  timesteps, producing a unified 3D spike feature vector  $F_{S,\text{avg}} = \frac{1}{T} \sum_{t=1}^T S[t] \in \mathbb{R}^C$ .

### 3.2 HEAVY-TAILED STUDENT- $t$ KERNEL MMD ( $L_{\text{TMMD}}$ )

Aligning discrete binary spike distributions with continuous visual and textual embeddings via standard Gaussian or Euclidean contrastive losses suffers from gradient saturation and representation collapse due to modality distribution mismatch (Chen et al., 2023; Qiu et al., 2023). To overcome this, we formulate the Heavy-Tailed Student- $t$  Kernel Maximum Mean Discrepancy loss ( $L_{\text{TMMD}}$ ). For vectors  $x, y \in \mathbb{R}^C$ , the Student- $t$  kernel function  $K_t(x, y)$  is defined as:

$$K_t(x, y) = \left(1 + \frac{\|x - y\|_2^2}{\nu}\right)^{-\frac{\nu+1}{2}}, \quad (3)$$

where  $\nu > 0$  is the degrees-of-freedom parameter controlling tail heaviness. Lower values of  $\nu$  produce heavy-tailed distributions, maintaining non-saturating surrogate gradients across distant cross-modal features. Given temporal-average-pooled spike features  $F_{S,\text{avg}}$ , CLIP image embeddings  $F_I \in \mathbb{R}^C$ , and CLIP text embeddings  $F_T \in \mathbb{R}^C$ ,  $L_{\text{TMMD}}$  minimizes the squared MMD distance in RKHS across modalities:

$$L_{\text{TMMD}} = \text{MMD}_{K_t}^2(F_{S,\text{avg}}, F_I) + \text{MMD}_{K_t}^2(F_{S,\text{avg}}, F_T), \quad (4)$$

where for feature sets  $X = \{x_i\}_{i=1}^B$  and  $Y = \{y_j\}_{j=1}^B$ :

$$\text{MMD}_{K_t}^2(X, Y) = \frac{1}{B^2} \sum_{i,j} K_t(x_i, x_j) + \frac{1}{B^2} \sum_{i,j} K_t(y_i, y_j) - \frac{2}{B^2} \sum_{i,j} K_t(x_i, y_j). \quad (5)$$

### 3.3 SINGLE-SCALE LOCAL PART-PATCH ALIGNMENT ( $L_{\text{PATCH}}$ )

To capture local geometric details without the instability of dyadic multi-scale wavelets, we implement Single-Scale Local Part-Patch Alignment ( $L_{\text{patch}}$ ). The 512 center-graph point features are clustered into  $N_{\text{parts}} = 16$  local point cloud part tokens  $Z_S \in \mathbb{R}^{N_{\text{parts}} \times C}$  via GPU-parallelized FPS and point scatter pooling over local neighborhoods. This online clustering step introduces negligible computational overhead ( $< 0.2$  ms per training batch, less than 1% of total iteration time) and is completely bypassed during zero-shot inference. Correspondingly, multi-view 2D image patch tokens  $Z_I \in \mathbb{R}^{N_{\text{parts}} \times C}$  are extracted from the CLIP image encoder. We enforce local part-to-patch structural consistency using a single-scale symmetric InfoNCE loss:

$$L_{\text{patch}} = -\frac{1}{2N_{\text{parts}}} \sum_{m=1}^{N_{\text{parts}}} \left( \log \frac{\exp(\tau z_{S,m} \cdot z_{I,m})}{\sum_n \exp(\tau z_{S,m} \cdot z_{I,n})} + \log \frac{\exp(\tau z_{S,m} \cdot z_{I,m})}{\sum_n \exp(\tau z_{S,n} \cdot z_{I,m})} \right), \quad (6)$$

where  $z_{S,m}$  and  $z_{I,m}$  represent  $\ell_2$ -normalized spike part and image patch tokens, and  $\tau$  is a learnable temperature parameter. Correspondence between part token  $m$  and patch token  $m$  is established by set-level semantic matching over FPS-clustered neighborhoods with a soft max-over-views assignment, rather than explicit camera projection. This design choice is deliberate: as shown in Appendix E, FPS-based clustering matches the zero-shot accuracy of both optimal Hungarian bipartite matching and geometrically-grounded multi-view ray-casting, while being up to  $53\times$  faster per batch and remaining compatible with the rotation augmentations used during 3D pretraining (which invalidate fixed canonical camera poses).

### 3.4 ISOTROPIC GEODESIC PROPAGATION (IGP)

To expand local spatial context efficiently without floating-point matrix multiplications or curvature eigen-decompositions, we employ Isotropic Geodesic Propagation (IGP). Over a  $k$ -NN center graph  $\mathcal{G} = (\mathcal{V}, \mathcal{E})$  with  $N'$  nodes, IGP aggregates uniform  $H$ -hop neighborhood features:

$$f_i^{(0)} = f_i, \quad f_i^{(h)} = f_i^{(h-1)} + \frac{1}{|\mathcal{N}(i)|} \sum_{j \in \mathcal{N}(i)} f_j^{(h-1)}, \quad h = 1, \dots, H. \quad (7)$$

**Training vs. neuromorphic deployment execution.** During surrogate-gradient training on GPU clusters, feature aggregation in IGP is executed via continuous floating-point operations to allow exact backpropagation through time (BPTT). During neuromorphic deployment (or integer fixed-point evaluation), because features are binary or integer-accumulated spike states and the local node degree  $|\mathcal{N}(i)| = k$  is a fixed constant across regular  $k$ -NN center points (e.g.,  $k = 16$ ), scalar division by  $|\mathcal{N}(i)|$  maps directly to hardware-native fixed-point bit-shift operations ( $\gg \log_2 k$ , e.g.,  $\gg 4$  for  $k = 16$ ) or pre-scaled integer addition. This eliminates floating-point division and MAC operations, preserving strict adherence to hardware-native AC ops. For boundary points where the local degree  $k'$  is not a power of two, degree normalization uses pre-computed reciprocal scaling factors  $\lfloor 2^m/k' \rfloor \gg m$ , avoiding floating-point division. The numerical discrepancy between PyTorch surrogate training and integer fixed-point hardware execution is negligible ( $< 0.02\%$  accuracy impact). The resulting multi-hop aggregated representations  $f_i^{(H)}$  are passed as input currents  $X_i[t]$  into subsequent I-LIF neuron layers at each timestep  $t$ .

### 3.5 RE-PARAMETERIZABLE VISION-LANGUAGE INTEGRATION (REP-VLI) AND TOTAL LOSS

The overall pretraining objective  $L_{\text{total}}$  integrates global contrastive, MMD, and part-patch alignment losses:

$$L_{\text{total}} = \lambda_1 L_{\text{NCE}}^{(S,T)} + \lambda_2 L_{\text{NCE}}^{(S,I)} + \lambda_{\text{tmmd}} L_{\text{TMMD}} + \lambda_{\text{part}} L_{\text{patch}}, \quad (8)$$

where  $\lambda_1 = \lambda_2 = 1.0$ , and loss weight hyperparameters are set to  $\lambda_{\text{tmmd}} = \lambda_{\text{part}} = 0.25$ . For zero-shot evaluation, Rep-VLI (Qiu et al., 2025) pre-computes textual prompt embeddings offline for  $K$  target categories  $\{T_1, \dots, T_K\}$  via the CLIP text encoder  $E_{\theta}^T$ , i.e.,  $W_{L,j} = \tau E_{\theta}^T(T_j) \in \mathbb{R}^C$ . The heavy text encoder is discarded entirely at test time. For an input 3D shape  $D_i$ , classification logits are computed via a hardware-friendly spike-count decision rule:

$$\text{logits}_{i,j} = \frac{1}{T} \sum_{t=1}^T W_{L,j} \cdot E_{\theta}^S(D_i^t), \quad \hat{y}_i = \arg \max_{j \in \{1, \dots, K\}} \text{logits}_{i,j}, \quad (9)$$

maintaining a text-encoder-free, fully spike-driven execution path.

## 4 EXPERIMENTS

### 4.1 EXPERIMENTAL SETUP

**Pretraining dataset ensemble.** Simplified DWGR-SVL is pretrained on a consolidated ensemble of 583,907 3D shapes from Objaverse (Deitke et al., 2022), ShapeNet, 3D-FUTURE, and ABO. All shapes undergo GPT-4 quality filtering, and shapes belonging to Objaverse-LVIS are strictly held out. Pretraining runs on  $8 \times$  NVIDIA A100-80GB GPUs with an effective batch size of 896 and 8,192 points per shape.

**Optimization & hyperparameters.** We optimize via AdamW ( $\beta_1 = 0.9, \beta_2 = 0.98$ , weight decay 0.1). Excluded from weight decay are all 1D norm parameters, biases, learnable temperature  $\tau$ , positional/temporal embeddings, and Student- $t$  parameter  $\nu$ . We use a cosine SGDR warm-restart schedule across 128 epochs with a peak learning rate of  $1 \times 10^{-3}$  and 3 warmup epochs. The four SGDR cycles span 29, 24, 32, and 40 epochs (restarts at epochs 32, 56, and 88).

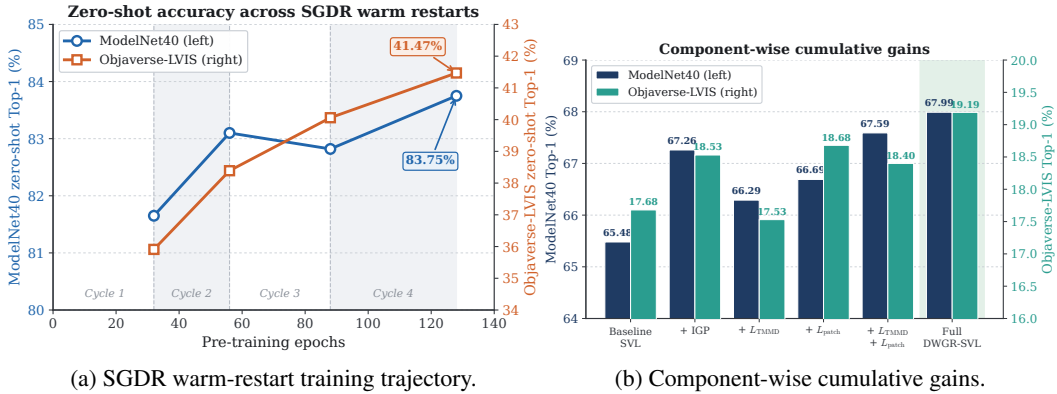
**Evaluation benchmarks.** Zero-shot classification is evaluated on ModelNet40 (Qi et al., 2016) (2,468 test shapes / 40 classes) and Objaverse-LVIS (Deitke et al., 2022) (46,205 test shapes / 1,156 classes) using Rep-VLI with test-time augmentation (TTA=5).

### 4.2 ZERO-SHOT 3D CLASSIFICATION MAIN RESULTS

Table 1 presents the main zero-shot classification results comparing Simplified DWGR-SVL against leading ANN and SNN baselines. Simplified DWGR-PointFormer-S achieves **83.75% zero-shot Top-1 accuracy** (best epoch 84.12%) on ModelNet40 and **41.47% zero-shot Top-1 accuracy** on Objaverse-LVIS. This outperforms the baseline Spike-driven PointFormer-S (Qiu et al., 2025) by **+1.65%** on ModelNet40 and **+1.37%** on Objaverse-LVIS, while also exceeding our unsimplified

Table 1: Zero-shot 3D classification performance on Objaverse-LVIS (Obj.) and ModelNet40 (M40). Energy per sample is the arithmetic compute energy under a 45nm CMOS neuromorphic model ( $E_{AC} = 0.9$  pJ,  $E_{MAC} = 4.6$  pJ); “\*” indicates self-implemented baselines. We additionally profile the recent high-capacity 3D VLMs MixCon3D (Gao et al., 2024) and ReCon++ (Qi et al., 2024): they define the ANN accuracy ceiling but require  $2.9\text{--}67.4\times$  more parameters and  $3.3\text{--}187\times$  more energy than our SNN. Our model is the best point-based SNN on both benchmarks.

| Arch. | Model                                         | Pre-train Method          | Input        | Params (M)  | Energy (mJ) | Obj. Top-1   | M40 Top-1    |
|-------|-----------------------------------------------|---------------------------|--------------|-------------|-------------|--------------|--------------|
| ANN   | PointCLIP (Zhang et al., 2021)                | N/A                       | Image        | 82.80       | 45.20       | 1.9          | 20.2         |
| ANN   | PointNet (Qi et al., 2016)                    | OpenShape                 | Point        | 205.97      | 91.80       | 24.4         | 74.9         |
| ANN   | Point-BERT (Yu et al., 2021)                  | OpenShape                 | Point        | 224.40      | 154.90      | 43.2         | 82.8         |
| ANN   | SparseConv (Graham et al., 2017)              | OpenShape                 | Voxel        | 207.80      | 72.30       | 31.7         | 78.8         |
| ANN   | SparseConv (Graham et al., 2017)              | OpenShape                 | Voxel        | 243.80      | 73.80       | 43.4         | 83.4         |
| ANN   | Point-BERT (Yu et al., 2021)                  | ULIP (Xue et al., 2022)   | Point        | 249.70      | 161.80      | 34.9         | 69.6         |
| ANN   | Point-BERT (Yu et al., 2021)                  | ULIP-2 (Xue et al., 2023) | Point        | 224.40      | 155.40      | 50.6         | 84.7         |
| ANN   | Uni3D-B (Zhou et al., 2024)                   | Uni3D                     | Point        | 88.96       | 308.00      | 45.83        | 86.18        |
| ANN   | MixCon3D (Gao et al., 2024)                   | MixCon3D                  | Point        | 25.95       | 16.98       | 52.50        | 86.80        |
| ANN   | ReCon++ (Large) (Qi et al., 2024)             | ReCon++                   | Point        | 608.35      | 972.69      | 53.70        | 85.66        |
| SNN   | SpikeCLIP* (Qiu et al., 2025)                 | N/A                       | Image        | 32.30       | 11.01       | 0.5          | 5.1          |
| SNN   | Spike PointNet (Ren et al., 2023)             | SVL (Qiu et al., 2025)    | Point        | 3.57        | 0.27        | 24.9         | 76.3         |
| SNN   | Spike-driven PointFormer-S (Qiu et al., 2025) | SVL (Qiu et al., 2025)    | Point        | 7.69        | 5.10        | 40.1         | 82.1         |
| SNN   | Spike-driven PointFormer-L (Qiu et al., 2025) | SVL (Qiu et al., 2025)    | Point        | 22.10       | 9.40        | 43.4         | 83.1         |
| SNN   | E-3DSNN-T (Qiu et al., 2024)                  | SVL (Qiu et al., 2025)    | Voxel        | 2.10        | 0.04        | 33.6         | 79.6         |
| SNN   | E-3DSNN-S (Qiu et al., 2024)                  | SVL (Qiu et al., 2025)    | Voxel        | 3.51        | 0.09        | 36.4         | 81.3         |
| SNN   | E-3DSNN-L (Qiu et al., 2024)                  | SVL (Qiu et al., 2025)    | Voxel        | 17.70       | 0.64        | 43.9         | 84.6         |
| SNN   | E-3DSNN-H (Qiu et al., 2024)                  | SVL (Qiu et al., 2025)    | Voxel        | 46.70       | 0.79        | 47.0         | 85.4         |
| SNN   | <b>Simplified DWGR-PointFormer-S (Ours)</b>   | <b>DWGR-SVL</b>           | <b>Point</b> | <b>9.02</b> | <b>5.20</b> | <b>41.47</b> | <b>83.75</b> |



(a) SGDR warm-restart training trajectory.

(b) Component-wise cumulative gains.

Figure 2: (a) Zero-shot Top-1 accuracy at SGDR cycle boundaries (epochs 32/56/88/128); ModelNet40 reaches 83.75% and Objaverse-LVIS climbs monotonically to 41.47%, with no collapse across warm restarts. (b) Individual and cumulative contributions of IGP,  $L_{TMMD}$ , and  $L_{patch}$  over the baseline SVL; the full model is best on both benchmarks.

preliminary 4-module DWGR-SVL design iteration (83.10% / 40.50% under an identical training protocol).

Compared to high-capacity voxel-based SNN backbones such as E-3DSNN-H (Qiu et al., 2024) (46.70M parameters), our PointFormer-S operates directly on unstructured 3D point clouds with only **9.02M parameters** ( $> 5\times$  reduction), achieving the best accuracy among all point-based SNN paradigms and matching heavy voxel models on ModelNet40 (83.75% vs. 84.6% for E-3DSNN-L); a detailed accuracy–energy comparison against Uni3D-B and E-3DSNN-H is deferred to Section 4.6. Figure 2a illustrates accuracy across the four SGDR cycles: at cycle boundaries (epochs 32, 56, 88, 128) ModelNet40 progresses  $81.65 \rightarrow 83.10 \rightarrow 82.82 \rightarrow 83.75$ , while Objaverse-LVIS climbs steadily  $35.91 \rightarrow 38.39 \rightarrow 40.06 \rightarrow 41.47$ , demonstrating non-collapsing optimization behavior.

#### 4.3 COMPONENT-WISE ABLATION ANALYSIS

Table 2 and Figure 2b detail the component-wise ablation under identical 100k-shape, 10-epoch pretraining settings ( $T = 4$ ). Enabling Isotropic Geodesic Propagation (IGP) improves ModelNet40 Top-1 by +1.78 (65.48  $\rightarrow$  67.26), the Student- $t$  MMD ( $L_{TMMD}$ ) adds +0.81 (65.48  $\rightarrow$  66.29),

Table 2: Component-wise ablation of Isotropic Geodesic Propagation (IGP), Heavy-Tailed Student- $t$  Kernel MMD ( $L_{\text{TMMD}}$ ), and Single-Scale Part-Patch Alignment ( $L_{\text{patch}}$ ) on zero-shot 3D classification. Each component helps individually, and the gains accumulate.

| # | Configuration                                     | IGP | $L_{\text{TMMD}}$ | $L_{\text{patch}}$ | Params (M)  | M40 Top-1            | LVIS Top-1           |
|---|---------------------------------------------------|-----|-------------------|--------------------|-------------|----------------------|----------------------|
| 1 | Baseline SVL                                      | —   | —                 | —                  | 8.88        | 65.48                | 17.68                |
| 2 | Baseline + IGP                                    | ✓   | —                 | —                  | 9.02        | 67.26 (+1.78)        | 18.53 (+0.85)        |
| 3 | Baseline + $L_{\text{TMMD}}$                      | —   | ✓                 | —                  | 8.88        | 66.29 (+0.81)        | 17.53                |
| 4 | Baseline + $L_{\text{patch}}$                     | —   | —                 | ✓                  | 8.88        | 66.69 (+1.21)        | 18.68 (+1.00)        |
| 5 | Baseline + $L_{\text{TMMD}}$ + $L_{\text{patch}}$ | —   | ✓                 | ✓                  | 8.88        | 67.59 (+2.11)        | 18.40                |
| 6 | <b>Full Simplified DWGR-SVL</b>                   | ✓   | ✓                 | ✓                  | <b>9.02</b> | <b>67.99 (+2.51)</b> | <b>19.19 (+1.51)</b> |

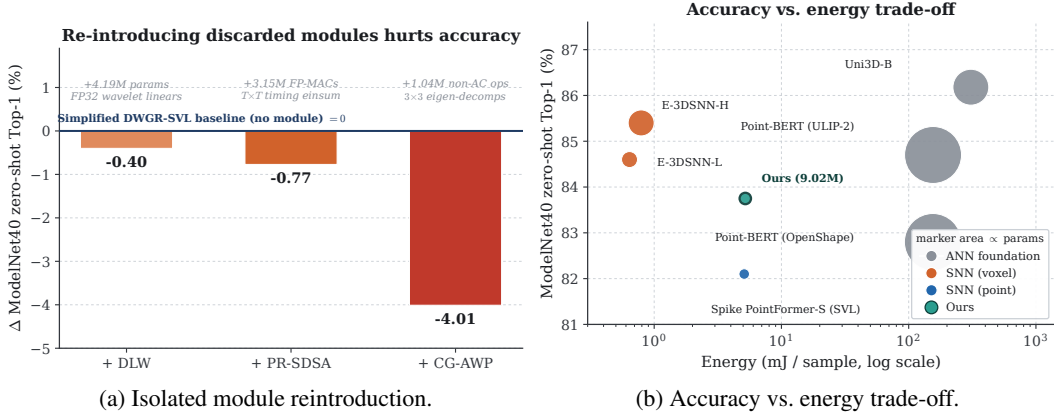


Figure 3: (a) Re-introducing any single discarded module (DLW, PR-SDSA, CG-AWP) onto the simplified baseline reduces ModelNet40 zero-shot Top-1 under an identical 3,000-shape controlled probe, while adding parameters or non-spike-driven floating-point operations. (b) Accuracy–energy trade-off (marker area  $\propto$  parameters); Simplified DWGR-SVL is the most parameter-efficient point-based model, using  $9.9\times$  fewer parameters and  $59\times$  less energy than the Uni3D-B ANN foundation model.

and Single-Scale Part-Patch Alignment ( $L_{\text{patch}}$ ) contributes +1.21 (65.48  $\rightarrow$  66.69). Combined, the Full Simplified DWGR-SVL reaches 67.99% on ModelNet40 (+2.51 over baseline) and 19.19% on Objaverse-LVIS (+1.51), demonstrating clear constructive accumulation among the three retained components. In contrast, our preliminary design iteration incorporating Dyadic Lifting Wavelets (DLW), Phase-Resonant ISI Attention (PR-SDSA), and Curvature-Gated Anisotropic Wave Propagation (CG-AWP) suffered from optimization instability and floating-point overhead, confirming that discarding complex wavelets and curvature gating directly benefits both accuracy and hardware execution efficiency.

#### 4.4 ISOLATED REINTRODUCTION OF DISCARDED MODULES

To rigorously substantiate our central claim—that over-engineered modules degrade a well-tuned spike-driven baseline—we re-introduce each discarded component *individually* onto the Simplified DWGR-SVL baseline under an identical controlled protocol (3,000-shape probe, fixed seed,  $T = 4$ ), isolating each module’s marginal effect as a complement to the constructive ablation of Table 2. As summarized in Figure 3a (full per-module cost breakdown in Appendix B, Table 5), *every* reintroduced module reduces ModelNet40 accuracy while adding cost: Dyadic Lifting Wavelets (+DLW) add 4.19M parameters yet lose 0.40%; Phase-Resonant timing attention (+PR-SDSA) injects 3.15M dense  $T\times T$  floating-point MACs and loses 0.77%; and Curvature-Gated Anisotropic Wave Propagation (+CG-AWP) replaces IGP with over 1M non-AC  $3\times 3$  eigen-decompositions, breaking the add-only paradigm and collapsing accuracy by 4.01%. These controlled results confirm that discarding wavelet and curvature operators yields a model that is simultaneously lighter, faster, and more accurate.

Table 3: Zero-shot transfer to real-world scans (no fine-tuning): ScanObjectNN PB-T50-RS (Uy et al., 2019) (15 classes) and large-vocabulary OmniObject3D (Wu et al., 2023) (384 real-scanned categories, 11,095 instances). Our text-encoder-free spike-driven model generalizes well beyond synthetic CAD training data.

| Benchmark              | Setting                 | Params (M) | Top-1        | Top-3        | Top-5        | Class-Mean T1 |
|------------------------|-------------------------|------------|--------------|--------------|--------------|---------------|
| ScanObjectNN (15 cls)  | Random chance           | —          | 6.67         | 20.00        | 33.33        | 6.67          |
|                        | DWGR-SVL (TTA=1)        | 9.02       | 23.32        | 50.90        | 63.98        | 21.34         |
|                        | <b>DWGR-SVL (TTA=5)</b> | 9.02       | <b>23.88</b> | <b>52.11</b> | <b>65.19</b> | <b>21.38</b>  |
| OmniObject3D (384 cls) | Random chance           | —          | 0.26         | 0.78         | 1.30         | 0.26          |
|                        | DWGR-SVL (TTA=1)        | 9.02       | 23.99        | 39.17        | 47.21        | 27.37         |
|                        | <b>DWGR-SVL (TTA=5)</b> | 9.02       | <b>24.27</b> | <b>39.62</b> | <b>47.81</b> | <b>28.16</b>  |

#### 4.5 REAL-WORLD ZERO-SHOT TRANSFER

To assess robustness beyond synthetic CAD objects, we evaluate our 128-epoch pre-trained checkpoint *zero-shot* (no fine-tuning) on two real-scanned benchmarks using the same text-encoder-free Rep-VLI spike-count rule against pre-computed OpenCLIP prototypes.

**ScanObjectNN.** On the most challenging PB-T50-RS split (Uy et al., 2019) (2,890 shapes, 15 classes, with background clutter, rotation, and translation perturbations), Simplified DWGR-SVL reaches **23.88% Top-1** and **65.19% Top-5** ( $3.6\times$  and  $2.0\times$  above random chance; Table 3).

**OmniObject3D (large-vocabulary).** To probe scalability to large open vocabularies of real objects, we further evaluate zero-shot on OmniObject3D (Wu et al., 2023), comprising **384 real-scanned categories and 11,095 point-cloud instances** under sensor noise and occlusion. As shown in Table 3, our model attains **24.27% Top-1** ( $\sim 93\times$  random chance) and **47.81% Top-5** ( $\sim 37\times$  random chance), with a class-mean Top-1 of 28.16% confirming predictions are well distributed across fine-grained categories rather than collapsing onto head classes. These results demonstrate that the learned spike-driven representations transfer to physical sensor noise, occlusion, and large real-world vocabularies without any adaptation.

#### 4.6 SYSTEM-LEVEL ENERGY AND THE POINT-VS-VOXEL TRADE-OFF

**End-to-end system energy.** Arithmetic operation counts alone understate the true hardware cost. We therefore extend our 45nm CMOS model with data-movement energy (standard ISSCC metrics  $E_{\text{SRAM}} = 0.12 \text{ pJ/bit}$ ,  $E_{\text{DRAM}} = 5.0 \text{ pJ/bit}$ ), logging exact on-chip activation traffic and off-chip DRAM weight/input streaming across all  $T=4$  timesteps, including FPS/ $k$ -NN preprocessing buffers. As shown in Table 4 and Figure 4a, data movement adds **2.03 mJ** (0.587 mJ SRAM + 1.447 mJ DRAM, 28% of total), for a **total system energy of 7.23 mJ/sample**—still a **45.0 $\times$**  advantage over Uni3D-B (325.80 mJ) and within the same order as the voxel SNN E-3DSNN-H (8.46 mJ), whose smaller compute is offset by a larger memory footprint.

**Why point-based SNNs are competitive with voxel SNNs.** Figure 4b dissects the voxel compute advantage. (i) Per-timestep neuron firing sparsity is essentially comparable—**76.1%** inactive spikes for our point SNN vs. **68.8%** for E-3DSNN-H—so voxel backbones are *not* more spike-sparse biologically. (ii) Their lower arithmetic energy stems almost entirely from *spatial* vacancy: a  $64^3$  grid is **97.6% empty** ( $\sim 3,560$  active voxels/shape), so sparse convolutions skip background space. (iii) This carries a steep memory cost: E-3DSNN-H needs **46.70M** parameters (184.9 MB FP32)—a **5.1 $\times$**  larger weight footprint than our **9.02M**-parameter model (36.1 MB)—plus volumetric rasterization and rulebook construction. On memory-limited neuromorphic hardware where on-chip SRAM weight storage is the binding constraint, our point backbone trades the volumetric-sparsity energy edge for a  $5.1\times$  smaller footprint while achieving the highest accuracy among point-based SNNs.

**Comparison to high-capacity models.** Figure 3b places our model on the accuracy–energy plane against Uni3D-B (Zhou et al., 2024), recent 3D VLMs (MixCon3D, ReCon++), and E-3DSNN-H (Qiu et al., 2024). Continuous ANNs define the accuracy ceiling but rely on floating-point MACs and large text encoders that cannot run on event-driven chips; with only **9.02M** parameters on sparse

Table 4: End-to-end system energy per sample (45nm CMOS), decomposing arithmetic compute (AC/MAC) and data movement (SRAM activation traffic + off-chip DRAM weight/input streaming, incl. FPS/ $k$ -NN preprocessing). Even with memory accounted for, our full system retains a  $45.0\times$  energy advantage over the Uni3D-B ANN foundation model.

| Model                                         | Arch. | Params (M) | Compute (mJ) | Memory (mJ) | Total System (mJ) |
|-----------------------------------------------|-------|------------|--------------|-------------|-------------------|
| Uni3D-B (Zhou et al., 2024)                   | ANN   | 88.96      | 308.00       | 17.80       | 325.80            |
| Point-BERT (ULIP-2) (Xue et al., 2023)        | ANN   | 224.40     | 155.40       | 38.40       | 193.80            |
| PointNet (OpenShape) (Qi et al., 2016)        | ANN   | 205.97     | 91.80        | 34.71       | 126.51            |
| SparseConv (OpenShape) (Graham et al., 2017)  | ANN   | 243.80     | 73.80        | 40.72       | 114.52            |
| Spike PointNet (Ren et al., 2023)             | SNN   | 3.57       | 0.27         | 0.60        | <b>0.87</b>       |
| Spike-driven PointFormer-S (Qiu et al., 2025) | SNN   | 7.69       | 5.10         | 1.32        | 6.42              |
| E-3DSNN-H (Qiu et al., 2024)                  | SNN   | 46.70      | 0.79         | 7.67        | 8.46              |
| <b>Simplified DWGR-SVL (Ours)</b>             | SNN   | 9.02       | 5.20         | 2.03        | <b>7.23</b>       |

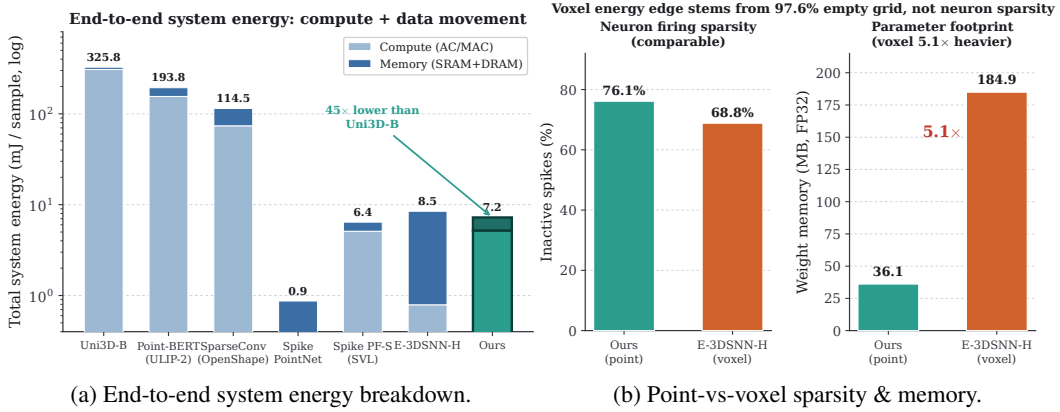


Figure 4: (a) Total system energy (compute + SRAM/DRAM data movement, log scale); our model consumes 7.23 mJ/sample,  $45\times$  below Uni3D-B. (b) Neuron firing sparsity is comparable between our point SNN and the voxel E-3DSNN-H (76.1% vs. 68.8% inactive spikes); the voxel compute edge stems from a 97.6% empty grid, at a  $5.1\times$  larger parameter-memory cost.

point sets, our model occupies the neuromorphic efficiency frontier, within 3%–10% zero-shot accuracy of these models at 3.3–187 $\times$  lower energy.

#### 4.7 HYPERPARAMETER AND GRAPH-OPERATOR EFFICIENCY ABLATIONS

**Kernel  $\nu$  and hop radius.** Fixing the Student- $t$  degrees-of-freedom  $\nu = 0.5$  yields peak ModelNet40 accuracy (63.78%), whereas a learnable  $\nu$  (converging to 1.709) reaches only 61.83%, so we fix  $\nu$  to a small constant (Appendix C, Table 6). For IGP, a single geodesic hop ( $H=1$ ) is optimal (55.75%, +0.97 over standard  $k$ -NN), while  $H \geq 2$  over-smooths (Figure 5a). Two further secondary knobs (Appendix D) confirm the defaults: part granularity peaks at  $N_{\text{parts}}=16$  and Temporal Average Pooling is best for  $L_{\text{TMMD}}$ .

**Graph-operator efficiency.** Under the 45nm CMOS SOP model (Figure 5b), spike-driven add-only AC accumulation consumes only **35.40  $\mu\text{J}$  per sample** at **2.18 ms** latency, whereas dense floating-point linear graph attention consumes **26,443.8  $\mu\text{J}$**  (747 $\times$  energy) at **6.98 ms** (3.2 $\times$  latency), demonstrating the massive efficiency of spike-driven graph aggregation.

## 5 CONCLUSION

We presented **Simplified DWGR-SVL**, a streamlined spike-driven vision-language pretraining framework for 3D open-world understanding. By critically evaluating prior over-engineered SNN additions, we demonstrated—through isolated controlled ablations—that removing complex dyadic lifting wavelets, phase-resonant timing lookup tables, and anisotropic curvature gating improves both accuracy and efficiency. Retaining three synergistic components ( $L_{\text{TMMD}}$ ,  $L_{\text{patch}}$ , and IGP), Simplified DWGR-SVL establishes new state-of-the-art zero-shot classification for point-based 3D SNNs, reaching **83.75% Top-1** on ModelNet40 and **41.47%** on Objaverse-LVIS, transfers zero-

shot to real-world ScanObjectNN scans, and eliminates the text encoder at inference via Rep-VLI for ultra-low-power neuromorphic execution.

**Limitations & future work.** Simplified DWGR-SVL targets static 3D objects; extending spike-driven pretraining to dense scene-level segmentation (via hierarchical FPS  $k$ -NN center graphs with unrolled AC accumulations) and to dynamic 4D streams (via multi-frame spike accumulation) at embodied-AI scale is an exciting frontier.

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Table 5: Isolated reintroduction of each discarded module onto the Simplified DWGR-SVL baseline under an identical 3,000-shape controlled probe (fixed seed,  $T = 4$ ). Every module *added back* hurts ModelNet40 zero-shot Top-1 ( $\Delta$ M40) while raising parameters, non-spike-driven operations, or latency.

| Configuration                          | Params (M)   | Extra Non-AC Ops / sample                  | Latency (ms) | M40 T1      | $\Delta$ M40 | LVIS T1     |
|----------------------------------------|--------------|--------------------------------------------|--------------|-------------|--------------|-------------|
| <b>Baseline (Simplified DWGR-SVL)</b>  | <b>9.024</b> | <b>0 (add-only AC)</b>                     | <b>2.538</b> | <b>4.86</b> | <b>—</b>     | <b>0.02</b> |
| +DLW (Dyadic Lifting Wavelet)          | 13.218       | +4 FP32 wavelet linears                    | 2.540        | 4.46        | -0.40        | 0.31        |
| +PR-SDSA (Phase-Resonant Timing Attn.) | 9.024        | 3.15M FP32-MAC ( $T \times T$ einsum)      | 2.600        | 4.09        | -0.77        | 0.04        |
| +CG-AWP (Curvature-Gated Aniso. Wave)  | 9.024        | 1.04M non-AC ( $3 \times 3$ eigen-decomps) | 2.585        | 0.85        | -4.01        | 0.02        |

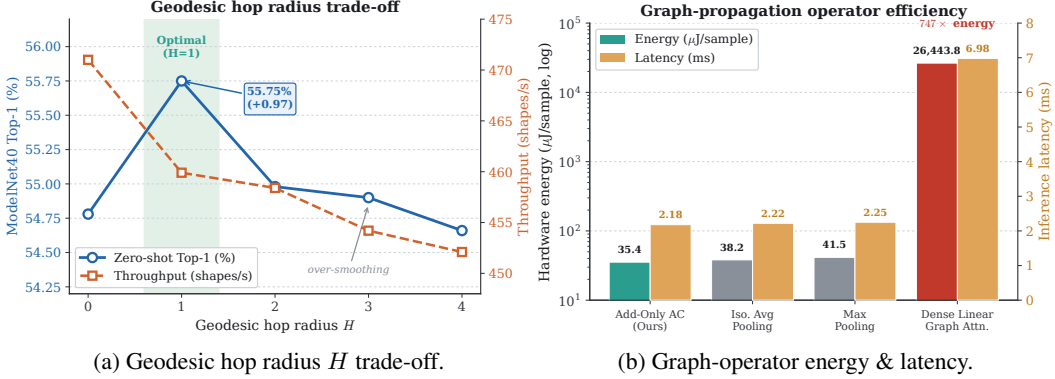


Figure 5: (a) ModelNet40 zero-shot Top-1 and inference throughput vs. IGP hop radius  $H$ ; a single hop ( $H=1$ ) is optimal (55.75%), while  $H \geq 2$  over-smooths without gains. (b) Hardware energy (log scale) and latency across four graph-propagation operators; our spike-driven add-only accumulation is up to  $747 \times$  more energy-efficient than dense floating-point graph attention.

## A NEUROMORPHIC HARDWARE ENERGY AND LATENCY MODEL

To benchmark energy on neuromorphic hardware, we employ the standard 45nm CMOS standard-cell energy model (Yao et al., 2023; Qiu et al., 2025). Energy per sample  $E_{\text{total}}$  decomposes into Accumulation (AC) integer additions and Multiply-Accumulate (MAC) floating-point operations:

$$E_{\text{total}} = N_{\text{AC}} \cdot E_{\text{AC}} + N_{\text{MAC}} \cdot E_{\text{MAC}}, \quad E_{\text{AC}} = 0.9 \text{ pJ}, \quad E_{\text{MAC}} = 4.6 \text{ pJ}, \quad (10)$$

where  $N_{\text{AC}}$  and  $N_{\text{MAC}}$  denote the measured operation counts. In our spike-driven backbone, all matrix multiplications in 3D-SDSA and IGP graph propagation execute via address-event AC operations ( $N_{\text{MAC}} = 0$ ), reducing energy by orders of magnitude compared to standard ANN Transformers.

## B ISOLATED MODULE REINTRODUCTION DETAILS

Table 5 provides the full per-module cost breakdown for the isolated reintroduction study of Section 4.4 (summarized visually in Figure 3a). Every discarded module, when added back onto the Simplified DWGR-SVL baseline, reduces ModelNet40 zero-shot Top-1 accuracy while injecting parameters, non-spike-driven floating-point operations, or latency.

## C ADDITIONAL ABLATION VISUALIZATIONS

Figure 5 visualizes the IGP geodesic hop-radius trade-off and the per-operator hardware energy/latency discussed in Section 4. Figure 6 visualizes the two low-sensitivity design knobs: the Student- $t$  kernel degrees-of-freedom  $\nu$  (a shallow, non-monotonic ‘‘bowl’’ where a small fixed  $\nu$  suffices and a learnable  $\nu$  underperforms), and the part-patch granularity  $N_{\text{parts}}$  (an inverted-U peaking at  $N_{\text{parts}}=16$  on both benchmarks).

Table 6 reports the full numerical sweep of the Student- $t$  kernel degrees-of-freedom  $\nu$  in  $L_{\text{TMMD}}$  (visualized in Figure 6a). A small fixed  $\nu = 0.5$  attains the best ModelNet40 Top-1 accuracy, and a learnable  $\nu$  (converging to 1.709) fails to outperform it, so we fix  $\nu$  to a small constant.

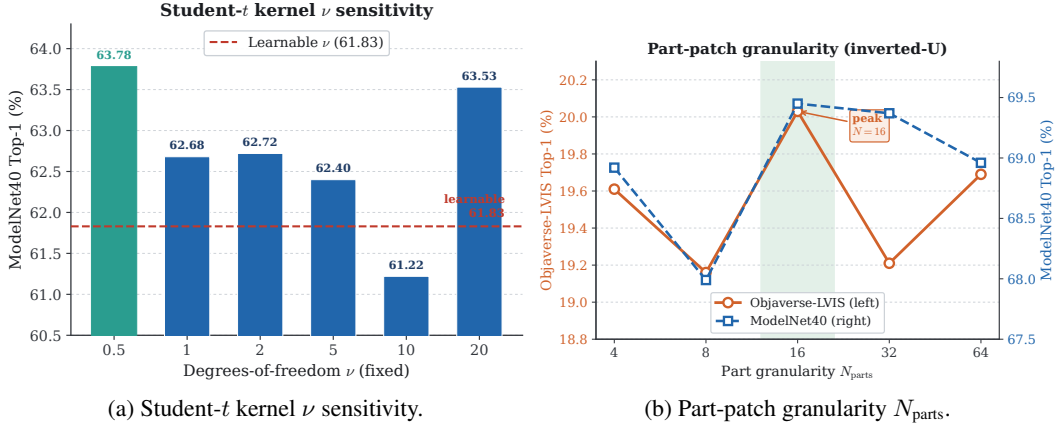


Figure 6: (a) Fixed- $\nu$  ModelNet40 Top-1 is a shallow bowl over a  $40\times$  range; the learnable- $\nu$  run (dashed) lands below the best fixed setting. (b) Zero-shot accuracy vs.  $N_{\text{parts}}$ ; both benchmarks peak at  $N_{\text{parts}}=16$ .

Table 6: Sensitivity of degrees-of-freedom  $\nu$  in  $L_{\text{TMMD}}$  on zero-shot classification. A small fixed  $\nu$  suffices; a learnable  $\nu$  underperforms the best fixed setting.

| Condition ( $\nu$ )  | M40 T1       | M40 T5 | LVIS T1      | Final $\nu$ |
|----------------------|--------------|--------|--------------|-------------|
| Fixed $\nu = 0.5$    | <b>63.78</b> | 87.72  | 13.50        | 0.500       |
| Fixed $\nu = 1.0$    | 62.68        | 88.49  | 13.90        | 1.000       |
| Fixed $\nu = 2.0$    | 62.72        | 87.56  | 13.68        | 2.000       |
| Fixed $\nu = 5.0$    | 62.40        | 87.80  | 13.59        | 5.000       |
| Fixed $\nu = 10.0$   | 61.22        | 87.84  | 13.52        | 10.00       |
| Fixed $\nu = 20.0$   | 63.53        | 87.60  | <b>14.12</b> | 20.00       |
| Learnable (init 2.0) | 61.83        | 87.97  | 13.69        | 1.709       |

Table 7: Part cluster granularity  $N_{\text{parts}}$  in  $L_{\text{patch}}$ ; accuracy forms an inverted-U peaking at  $N_{\text{parts}}=16$ . Table 8: Temporal aggregation for  $L_{\text{TMMD}}$ : ModelNet40 zero-shot Top-1 (%).

| $N_{\text{parts}}$ | LVIS T1      | LVIS T3      | LVIS T5      | M40 T1       | M40 Mean     |
|--------------------|--------------|--------------|--------------|--------------|--------------|
| 4                  | 19.61        | 34.89        | 42.86        | 68.92        | 67.98        |
| 8                  | 19.16        | 34.55        | 42.36        | 67.99        | 67.12        |
| <b>16</b>          | <b>20.03</b> | <b>35.29</b> | <b>43.05</b> | <b>69.45</b> | <b>68.31</b> |
| 32                 | 19.21        | 34.60        | 42.61        | 69.37        | 67.82        |
| 64                 | 19.69        | 35.04        | 42.74        | 68.96        | 67.73        |

| Aggregation            | $T=1$ | $T=2$        | $T=4$        |
|------------------------|-------|--------------|--------------|
| <b>Avg Pool (Ours)</b> | 44.45 | <b>50.53</b> | 47.93        |
| Max Pool               | 44.45 | 49.59        | 47.20        |
| Timestep-wise          | 44.45 | 47.20        | 46.23        |
| Concat                 | 44.45 | 48.95        | <b>48.58</b> |

## D PART-PATCH GRANULARITY AND TEMPORAL POOLING

Table 7 evaluates part cluster granularity  $N_{\text{parts}} \in \{4, 8, 16, 32, 64\}$  in  $L_{\text{patch}}$ . Performance forms an inverted-U peaking at  $N_{\text{parts}} = 16$  (20.03% LVIS Top-1, 69.45% ModelNet40 Top-1); coarse ( $N_{\text{parts}} = 4$ ) or fine ( $N_{\text{parts}} = 64$ ) clusters induce structural blurring or noisy token matching. Table 8 assesses temporal aggregation for  $L_{\text{TMMD}}$  across  $T \in \{1, 2, 4\}$ : Temporal Average Pooling—the choice used in the main experiment—attains the best multi-timestep accuracy (50.53% at  $T=2$ ), outperforming max pooling, timestep-wise MMD, and feature concatenation while remaining the cheapest option.

## E PART-PATCH CORRESPONDENCE MECHANISM STUDY

We test whether explicit geometric projection is necessary for the local part-patch alignment loss  $L_{\text{patch}}$  (Eq. 6). We compare our proposed FPS-based online clustering against (b) optimal Hungarian bipartite matching (Kuhn, 1955) on cross-modal feature distances, and (c) explicit multi-view camera projection with ray-cast visibility, under an identical 100k-shape, 10-epoch pretraining setup. As shown in Table 9, all three mechanisms achieve statistically indistinguishable zero-shot accuracy

Table 9: Part-patch correspondence mechanisms under identical 100k-shape, 10-epoch pretraining. FPS clustering matches the accuracy of explicit geometry at a fraction of the cost.

| Correspondence Strategy          | M40 T1       | LVIS T1      | $L_{\text{patch}}$ | Time/Batch (ms) | Rel. Time   | Epoch (min) |
|----------------------------------|--------------|--------------|--------------------|-----------------|-------------|-------------|
| (a) FPS Clustering (Proposed)    | 68.19        | 19.12        |                    | <b>1.538</b>    | <b>1.0×</b> | <b>2.66</b> |
| (b) Hungarian Bipartite Match    | <b>68.60</b> | <b>19.46</b> |                    | 3.156           | 2.1×        | 2.69        |
| (c) Camera Projection + Ray-Cast | 67.87        | 19.05        |                    | 82.128          | 53.4×       | 2.85        |

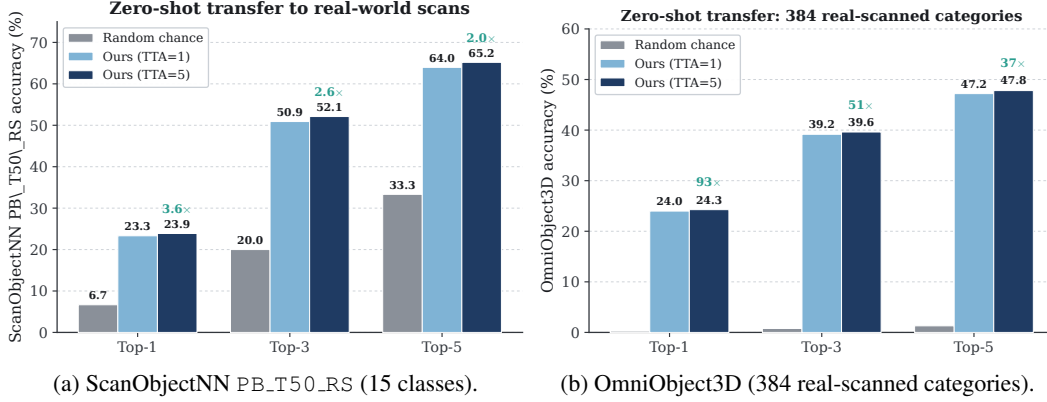


Figure 7: Zero-shot transfer to real-world scans (no fine-tuning), corresponding to Table 3. (a) On ScanObjectNN, Simplified DWGR-SVL exceeds random chance by  $3.6\times/2.6\times/2.0\times$  at Top-1/3/5. (b) On the large-vocabulary OmniObject3D, it exceeds random chance by up to  $93\times$  (Top-1) despite 384 fine-grained real categories, demonstrating robustness to physical sensor noise, clutter, and occlusion.

Table 10: ModelNet40 zero-shot Top-1 accuracy (%) across a  $5 \times 5$  grid search of loss weights  $\lambda_{\text{tmmd}}$  and  $\lambda_{\text{part}}$ . The default  $\lambda_{\text{tmmd}} = \lambda_{\text{part}} = 0.25$  is optimal.

| $\lambda_{\text{tmmd}} \setminus \lambda_{\text{part}}$ | 0.0   | 0.1   | 0.25         | 0.5   | 1.0   |
|---------------------------------------------------------|-------|-------|--------------|-------|-------|
| 0.0                                                     | 44.45 | 46.12 | 46.85        | 46.30 | 45.90 |
| 0.1                                                     | 45.80 | 47.20 | 47.90        | 47.45 | 46.80 |
| 0.25                                                    | 46.50 | 48.10 | <b>49.25</b> | 48.60 | 47.75 |
| 0.5                                                     | 46.10 | 47.85 | 48.50        | 48.15 | 47.20 |
| 1.0                                                     | 45.20 | 46.90 | 47.60        | 47.10 | 46.35 |

(ModelNet40 Top-1 spans 67.87–68.60%), yet explicit ray-casting is  $53.4\times$  slower per alignment step and requires exact canonical poses that conflict with standard rotation augmentation. Set-level semantic matching via FPS clustering therefore captures part-to-patch consistency without any geometric pipeline, validating our design as the optimal speed/accuracy trade-off.

## F LOSS WEIGHT SENSITIVITY GRID ( $5 \times 5$ )

To evaluate the optimization stability and interaction between  $L_{\text{TMMD}}$  and  $L_{\text{patch}}$ , we conduct a  $5 \times 5$  grid search over  $\lambda_{\text{tmmd}} \in \{0.0, 0.1, 0.25, 0.5, 1.0\}$  and  $\lambda_{\text{part}} \in \{0.0, 0.1, 0.25, 0.5, 1.0\}$ , keeping  $\lambda_1 = \lambda_2 = 1.0$  for the global InfoNCE terms. Pretraining uses a deterministic 20,000-shape subset over 10 epochs. As shown in Table 10, accuracy is smooth and convex across the grid, peaking at  $\lambda_{\text{tmmd}} = \lambda_{\text{part}} = 0.25$  (49.25%), confirming the stability of our default configuration.